

Chapter 05 Part 1: Mass Spec and Moles

1.

Compound	Molar Mass (grams)
Na ₂ O	62.0
MgO	40.3
K ₂ O	94.2
CaO	56.1

According to the information in the table above, a 1.00 g sample of which of the following contains the greatest mass of oxygen?

(A) Na₂O

(B) MgO

(C) K₂O

(D) CaO



2. In his atomic theory, Dalton proposed that all atoms of a given element are identical. Which of the following observations provides the best evidence that the proposal is incorrect?

(A) The mass spectrum of Cu has a peak at 63 amu and another peak at 65 amu.

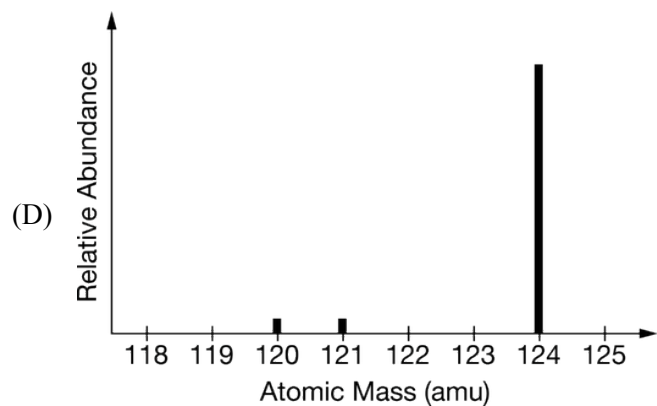
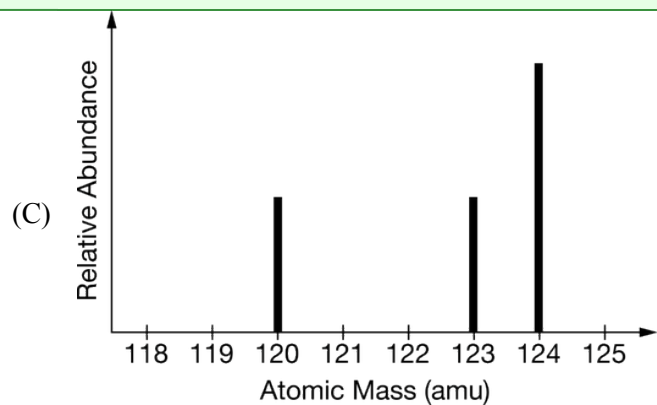
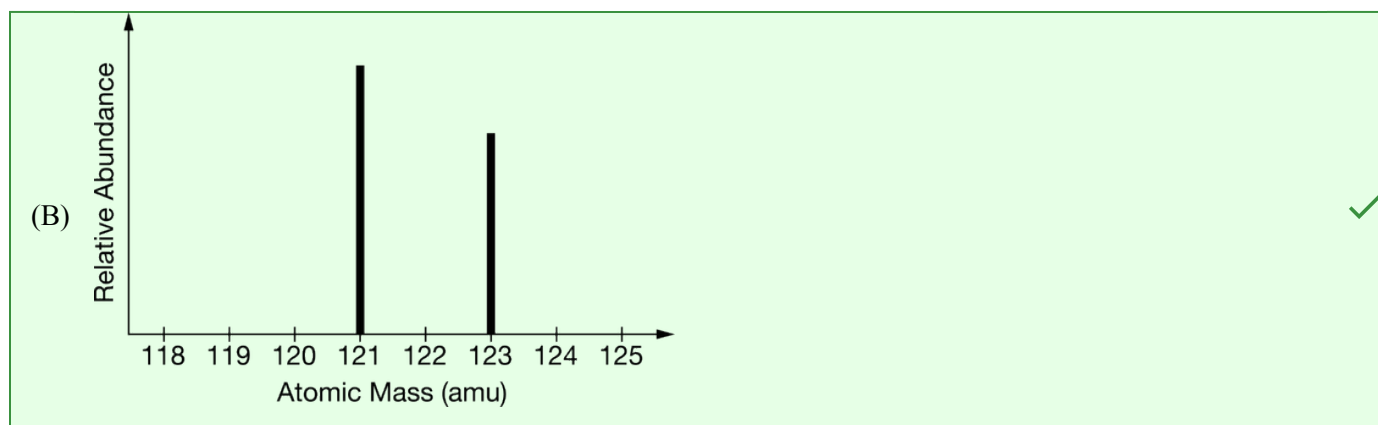
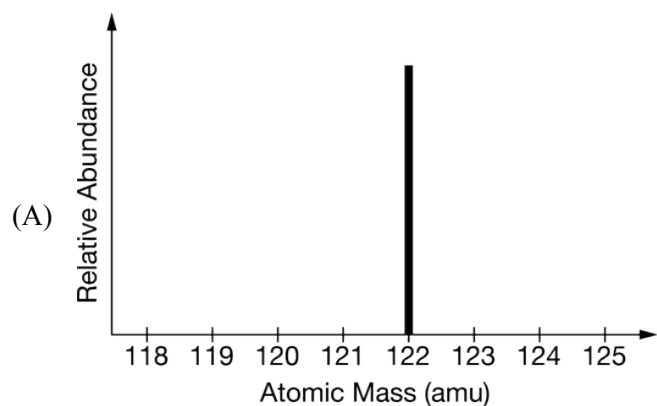
(B) The ionic radius of Cl⁻ is 0.181 nm, and the atomic radius of Cl is 0.079 nm.(C) The dipole moment of O₃ is 0.53 debye, and the dipole moment of O₂ is 0.00 debye.

(D) Elemental P can exist in both white and red forms.



3. The mass spectrum of the element Sb is most likely represented by which of the following?

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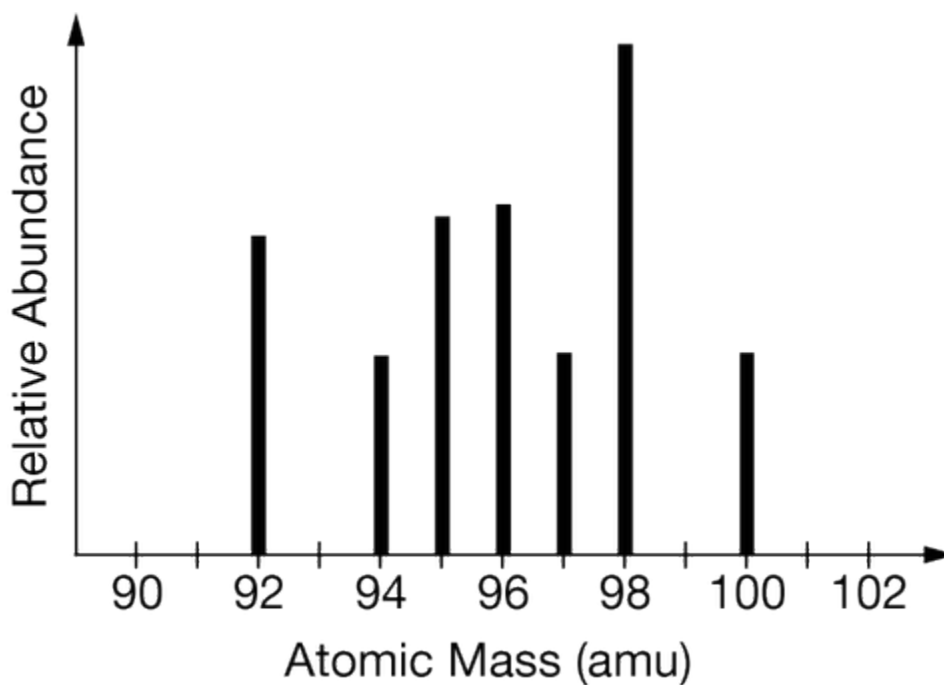
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4. Given that the density of $\text{Hg}(l)$ at 0°C is about 14 g mL^{-1} , which of the following is closest to the volume of one mole of $\text{Hg}(l)$ at this temperature?

(A) 0.070 mL
(B) 0.14 mL
(C) 1.4 mL
(D) 14 mL
(E) 28 mL



5.



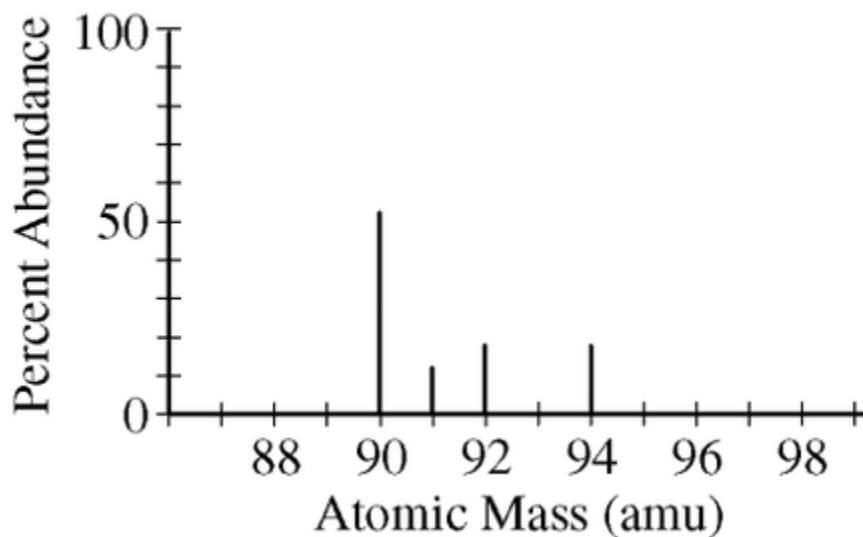
Which of the following elements has the mass spectrum represented above?

(A) Nb
(B) Mo
(C) U
(D) Cf



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6.



The mass spectrum of an average sample of a pure element is shown in the figure above. Which of the following is the identity of the element?

(A) Y

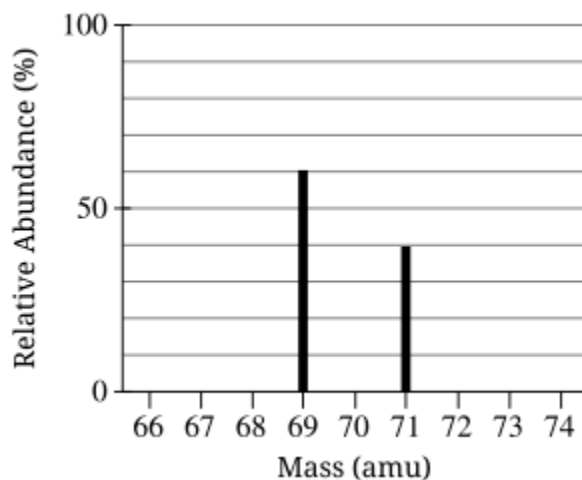
(B) Zr

(C) Nb

(D) Th



7.



The mass spectrum of an element is represented above. Which of the following is most likely the identity of the element?

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(A) Ga

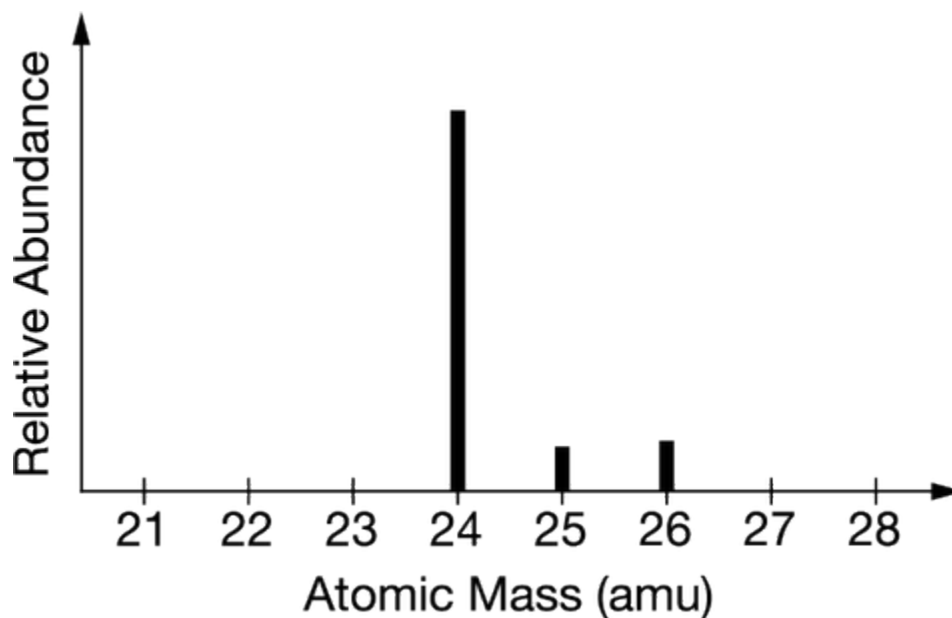


(B) Ge

(C) Tm

(D) Lu

8.



The mass spectrum of a sample of a pure element is shown above. Based on the data, the peak at 26 amu represents an isotope of which of the following elements?

(A) Al with 13 neutrons

(B) Mg with 14 neutrons



(C) Fe with 26 neutrons

(D) Ti with 26 neutrons

9. A student has a 1 g sample of each of the following compounds: NaCl, KBr, and KCl. Which of the following lists the samples in order of increasing number of moles in the sample?

(A) NaCl < KCl < KBr

(B) NaCl < KBr < KCl

(C) KCl < NaCl < KBr

(D) KBr < KCl < NaCl

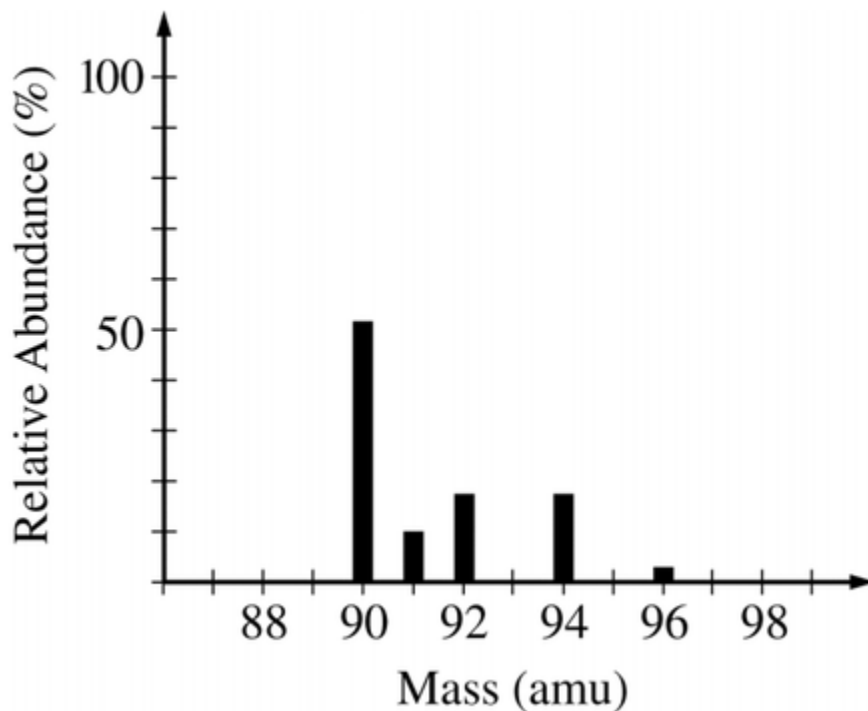


10. A student obtains a sample of a pure solid compound. In addition to Avogadro's number, which of the following must the student know in order to determine how many molecules are in the sample?

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- (A) Mass of the sample, volume of the sample
(B) Mass of the sample, density of the sample
(C) Molar mass of the compound, mass of the sample ✓
(D) Molar mass of the compound, density of the sample

11.



The mass spectrum of element X is presented in the diagram above. Based on the spectrum, which of the following can be concluded about element X?

- (A) X is a transition metal, and each peak represents an oxidation state of the metal.
(B) X contains five electron sublevels.
(C) The atomic mass of X is 90.
(D) The atomic mass of X is between 90 and 92. ✓

12. A solution of methanol, CH_3OH , in water is prepared by mixing together 128 g of methanol and 108 g of water. The mole fraction of methanol in the solution is closest to

- (A) 0.80
(B) 0.60
(C) 0.50
(D) 0.40 ✓
(E) 0.20

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13. A solution is prepared by adding 16 g of CH_3OH (molar mass 32 g) to 90. g of H_2O (molar mass 18 g). The mole fraction of CH_3OH in this solution is closest to which of the following?

(A) 0.1



(B) 0.2

(C) 0.3

(D) 0.4

(E) 0.6

14. In 1.00 mol of potassium zirconium sulfate trihydrate, $\text{K}_4\text{Zr}(\text{SO}_4)_4 \cdot 3 \text{H}_2\text{O}$, there are

(A) $3 \times 6.02 \times 10^{23}$ hydrogen atoms

(B) 6.02×10^{23} sulfur atoms

(C) $4 \times 6.02 \times 10^{23}$ potassium atoms



(D) 4 moles of oxygen atoms

(E) 4 moles of zirconium atoms

15. How many carbon atoms are contained in 2.8 g of C_2H_4 ?

(A) 1.2×10^{23}



(B) 3.0×10^{23}

(C) 6.0×10^{23}

(D) 1.2×10^{24}

(E) 6.0×10^{24}

16. A 100.0 g sample of which of the following compounds contains the most molecules?

(A) CO_2

(B) SO_2

(C) H_2O



(D) N_2O

17. Which of the following numerical expressions gives the number of particles in 2.0 g of Ne?

(A) $\frac{6.0 \times 10^{23} \text{ particles/mol}}{2.0 \text{ g}}$

(B) $\frac{6.0 \times 10^{23} \text{ particles/mol}}{20.18 \text{ g/mol}}$

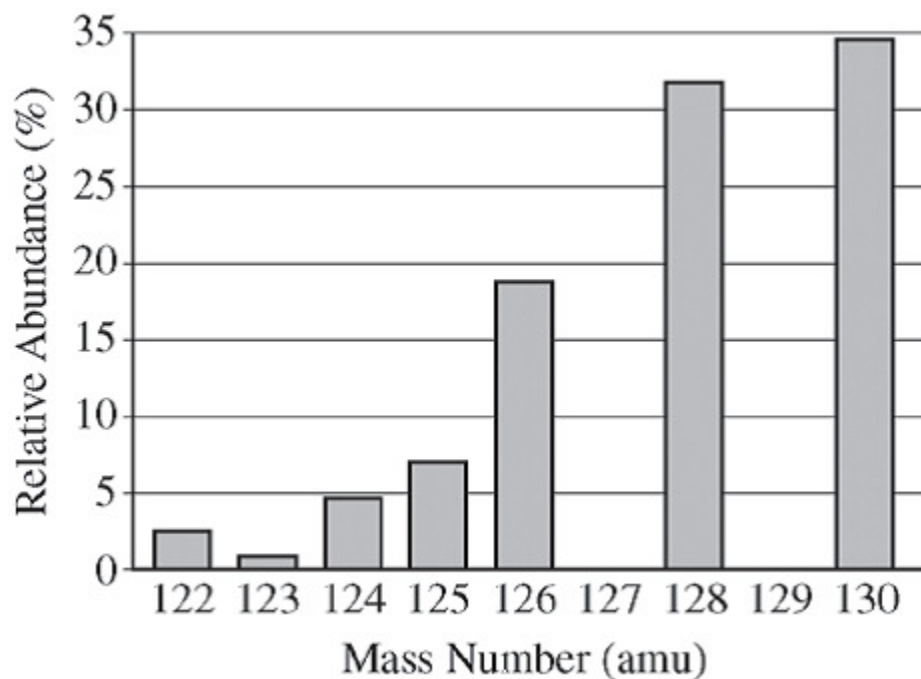
(C) $\frac{2.0 \text{ g}}{20.18 \text{ g/mol}} (6.0 \times 10^{23} \text{ particles/mol})$



(D) $\frac{20.18 \text{ g/mol}}{2.0 \text{ g}} (6.0 \times 10^{23} \text{ particles/mol})$

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18.



The elements I and Te have similar average atomic masses. A sample that was believed to be a mixture of I and Te was run through a mass spectrometer, resulting in the data above. All of the following statements are true. Which one would be the best basis for concluding that the sample was pure Te?

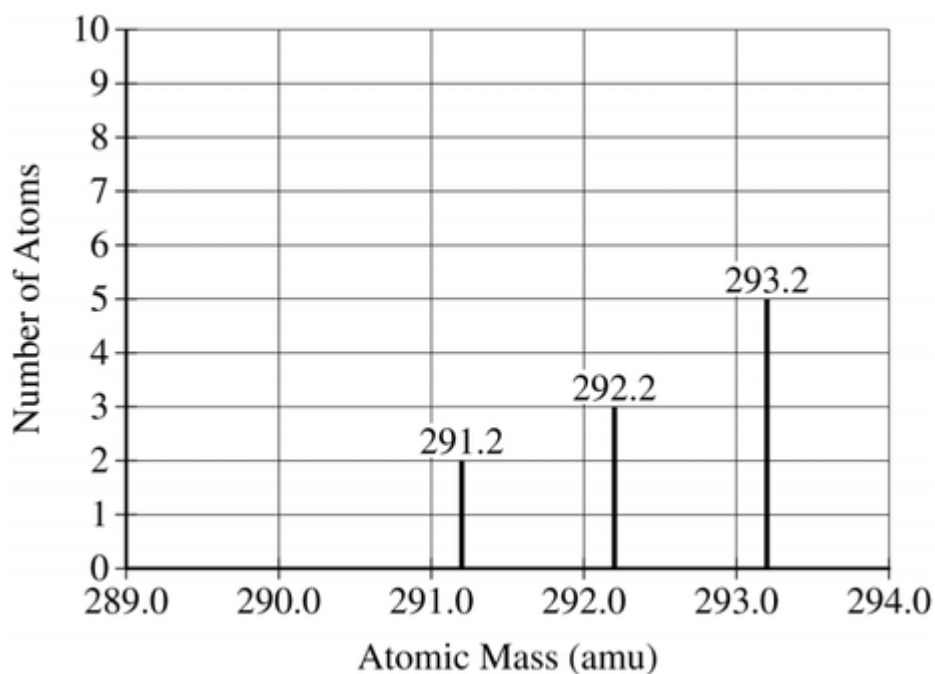
- (A) Te forms ions with a -2 charge, whereas I forms ions with a -1 charge.
- (B) Te is more abundant than I in the universe.

(C) I consists of only one naturally occurring isotope with 74 neutrons, whereas Te has more than one isotope. ✓

- (D) I has a higher first ionization energy than Te does.

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19. A new element with atomic number 116 was discovered in 2000. In 2012 it was named livermorium, Lv. Although Lv is radioactive and short-lived, its chemical properties and reactivity should follow periodic trends.
- Write the electron configuration for the valence electrons of Lv in the ground state.
 - According to periodic properties, what would be the most likely formula for the product obtained when Lv reacts with $\text{H}_2(\text{g})$?
 - The first ionization energy of polonium, Po, is 812 kJ/mol. Is the first ionization energy of Lv expected to be greater than, less than, or equal to that of Po? Justify your answer in terms of Coulomb's law.
 - Shown below is a hypothetical mass spectrum for a sample of Lv containing 10 atoms.



Using the information in the graph, determine the average atomic mass of Lv in the sample to four significant figures.

Part A

1 point(s) maximum

1 point is earned for the correct configuration.

7s² 7p⁴



0

1

Chapter 05 Part 1: Mass Spec and Moles**Student response earns 1 of the following 1 point(s)****1 point(s) maximum**

1 point is earned for the correct configuration.

7s² 7p⁴**Part B****1 point(s) maximum**

1 point is earned for the correct formula.

LvH₂ (or H₂Lv)

0	1
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Student response earns 1 of the following 1 point(s)**1 point(s) maximum**

1 point is earned for the correct formula.

LvH₂ (or H₂Lv)**Part C****1 point(s) maximum**

1 point is earned for a correct prediction with a valid justification.

Less than that of Po.

The two atoms have comparable effective nuclear charges, but the valence electrons in Lv would be at a greater distance from the nucleus than those in Po. By Coulomb's law, the attractive force between the valence electrons and the nucleus decreases by the inverse square of the distance between them.



0	1
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Student response earns 1 of the following 1 point(s)**1 point(s) maximum**

1 point is earned for a correct prediction with a valid justification.

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Less than that of Po.

The two atoms have comparable effective nuclear charges, but the valence electrons in Lv would be at a greater distance from the nucleus than those in Po. By Coulomb's law, the attractive force between the valence electrons and the nucleus decreases by the inverse square of the distance between them.

Part D**1 point(s) maximum**

1 point is earned for the correct average atomic mass.

Average atomic mass =

$$\frac{2}{10}(291.2) + \frac{3}{10}(292.2) + \frac{5}{10}(293.2) = 292.5 \text{ amu}$$



0	1
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Student response earns 1 of the following 1 point(s)

1 point(s) maximum

1 point is earned for the correct average atomic mass.

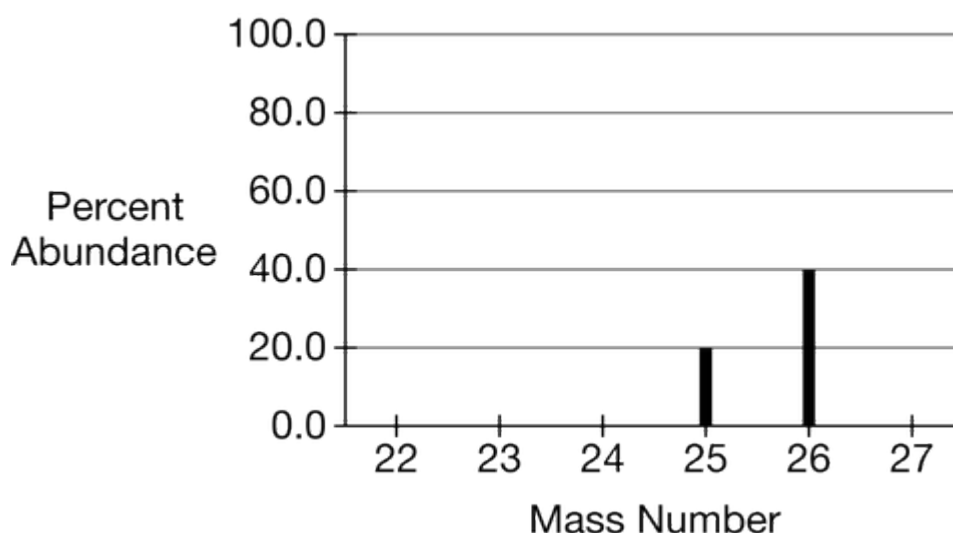
Average atomic mass =

$$\frac{2}{10}(291.2) + \frac{3}{10}(292.2) + \frac{5}{10}(293.2) = 292.5 \text{ amu}$$

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20. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

The three stable isotopes of Mg are Mg-24, Mg-25, and Mg-26. A rock sample from another planet has a different distribution of the three stable isotopes than what is found on Earth. A partial mass spectrum for Mg in the rock sample is shown below.



- (a) Draw a line on the graph to show the mass of the missing isotope and its percent abundance.
- (b) Use the completed mass spectrum to calculate the average atomic mass of Mg on the other planet.
- (c) The first ionization energy of magnesium samples found on Earth is experimentally determined to be 736 kJ/mol. If a similar experiment is conducted to determine the first ionization energy of magnesium in the rock sample from another planet, will it be greater than, less than, or equal to the first ionization energy of magnesium found on Earth? Explain your reasoning.

	Successive Ionization Energies (kJ/mol)		
	First	Second	Third
Mg	736	1450	7740
Unknown element	496	4560	6910

- (d) A table of successive ionization energies for Mg and an unknown element is shown above. Predict the identity

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of the unknown element.

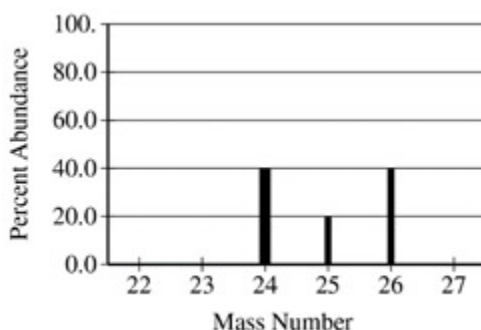
Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct a correct answer showing a line with a height of 40 and an atomic mass of 24 amu .

**Part B**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct calculated value.

$$\frac{24(40)+25(20)+26(40)}{100} = 25 \text{ amu}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a correct answer and a valid explanation equivalent to one of the following.

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· Equal to. All atoms of magnesium have the same number of protons and electrons and therefore the same first ionization energy.

OR

· Equal to. Ionization energy is not affected by the number of neutrons (neutral particles) present and is therefore identical for all isotopes.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a correct answer.

· Sodium (or any alkali metal)